

Regional Favoritism in Later Periods: Nightlights and Pollution in Leader Birth Regions

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Compiled on May 1, 2026

We study how the canonical birth-region favoritism result in nightlights evolves over time. We combine the Political Leaders' Affiliation Database, ADM2-level nightlights, ACAG surface NO₂, ACAG PM_{2.5}, V-Dem democracy scores, and GADM administrative boundaries in a global district-year panel with region fixed effects and country-year fixed effects. In the long DMSP window, the lag-0 birth-region coefficient is positive at 0.0128 in 1992–2013 ($p = 0.030$). But 5-year interactions within DMSP show a sharper temporal pattern: the effect is strongest in 2007–2011 at 0.0292 ($p = 0.004$) and then falls to -0.0057 in 2012–2013 ($p = 0.709$), so the break begins before any sensor switch. The historical positive effect is concentrated in Sub-Saharan Africa, where the 1992–2013 coefficient is 0.0442 ($p = 0.011$), while non-SSA is close to zero. Yet no post-2005 subsample restores a precise positive activity effect. A standalone VIIRS-only panel for 2014–2024 also yields negative later-period estimates. Pollution outcomes do not support the idea that leaders shifted from directing more growth to directing cleaner growth toward birth regions. The main contribution is to document a late-2000s rise and early-2010s break in the activity effect rather than a simple gradual fade.

regional favoritism | political economy | nighttime lights | air pollution | autocracy

Political leaders often direct resources toward their home regions, a pattern documented in cross-country evidence using subnational nightlights (1). This paper asks whether that result still holds in later periods, and how it changes over time inside the original DMSP era. The post-2005 years are especially useful for that question because they overlap with harmonized nightlights and global pollution measures. They are also exactly where the canonical birth-region activity effect becomes much harder to detect.

The paper's central empirical result is not a smooth attenuation. Within DMSP, the birth-region effect is modest in early 5-year bins, strongest in 2007–2011, and then much weaker by 2012–2013. That break starts before the move to later nightlights products. The historical positive effect is also geographically concentrated: Sub-Saharan Africa drives the long-window replication, while non-SSA estimates are much smaller. In later annual windows, however, no targeted post-2005 subsample restores a precise positive activity effect.

Pollution outcomes enter as a diagnostic for one possible explanation. If leaders still favor birth regions after 2005, but do so through the composition rather than the quantity of growth, then birth regions may receive cleaner activity even when nightlights do not rise. We define “green favoritism” as lower pollution per unit of proxied economic activity in leader birth regions. This possibility matters for environmental inequality as well as distributive politics (2). The pollution evidence does not support it: NO₂ and PM_{2.5} estimates are zero to positive rather than negative.

The contribution relative to recent replication work is there-

fore diagnostic. Recent work with expanded birthplace data and harmonized nightlights also finds a positive pooled favoritism effect in extended samples (3, 4). By contrast, we focus on the temporal pattern inside later annual windows, show that the break starts within DMSP before any sensor switch, isolate the long-run Sub-Saharan African historical core, and evaluate whether a shift toward cleaner growth can explain the weaker activity effect. The result is not that the historical finding disappears in all data. It is that the later-period relationship is unstable: it peaks in the late 2000s, weakens sharply by the early 2010s, and is not restored by post-2005 subsample cuts or by standalone VIIRS data.

1. Data and Empirical Design

The analysis combines five core inputs. First, the Political Leaders' Affiliation Database (PLAD) provides leader identity, spell timing, and subnational birthplace information used to code treatment (5). Second, nightlights provide the economic-activity proxy. We use a replication-style DMSP ADM2 panel for 1992–2013, a harmonized DMSP-to-VIIRS panel for 2005–2019, and a standalone VIIRS ADM2 panel for 2012–2024 as a single-sensor later diagnostic. DMSP records luminance as an integer Digital Number (DN) from 0 to 63, with known saturation near the ceiling (6). Third, we use ACAG surface NO₂ data for the 2005–2019 pollution window (7). Fourth, we use ACAG SatPM PM_{2.5} as a longer-window pollution extension. Fifth, GADM ADM2 boundaries provide the common subnational geography (8).

Treatment is defined at the district-year level. A district is treated when it is the birth region of the national leader in office in that year. To ensure a unique treatment assignment in transition years, we truncate the earlier of any two consecutive leader spells that overlap within a country.

Significance Statement

Regional favoritism is a well-known finding in political economy: leaders tend to direct economic activity toward their birth regions. We show that this canonical nightlights result does not simply fade gradually. Within the DMSP era, it is strongest in the late 2000s and then weakens sharply by 2012–2013, before any switch to later satellite products. The historical positive effect is concentrated in Sub-Saharan Africa, but no post-2005 subsample or pollution-composition test restores it.

R.S. conceived the study, built and integrated the main data pipeline, ran the core empirical analysis, and drafted the manuscript. M.M.A. contributed to the NO₂ analysis. R.W. contributed to manuscript review and proofreading.

The authors declare no competing interests.

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The core estimating equation uses ADM2 fixed effects and country-year fixed effects:

$$\log(Y_{ict} + 0.01) = \beta \text{BirthRegionLeader}_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict},$$

where i indexes ADM2 regions, c countries, and t years. With ADM2 and country-year fixed effects, β is identified by changes in the current leader's birth district relative to other districts in the same country-year, net of permanent district differences. A causal interpretation requires birth and non-birth districts to have followed parallel within-country trends absent leader ascension. That assumption could fail if future leaders come from districts already on different trajectories or if national transitions coincide with region-specific shocks, so we interpret the estimates as reduced-form evidence rather than a fully causal decomposition.

Standard errors are clustered by leader spell in the baseline, following the leader-term clustering convention in recent replication work such as (3). Because (1) and (4) use country-level clustering, we also report country-clustered robustness checks for the main estimates. Following the baseline specification in (1), we estimate lag 0, lag 1, and lag 2 treatment indicators, and focus on lag 0 in the main text as the most direct test of the regional favoritism hypothesis.

Our later-period environmental outcomes are log nightlights, log NO₂, and a pollution-intensity measure defined as $\log(\text{NO}_2 + 0.01) - \log(\text{Nightlights} + 0.01)$. Because both component regressions use the same regressor and fixed effects, the intensity coefficient is mechanically a linear combination of the NO₂ and nightlights coefficients ($\beta_{\text{intensity}} = \beta_{\text{NO}_2} - \beta_{\text{nightlights}}$). We report this measure as a compact summary of whether pollution rises more or less than nightlights in birth regions. Because it is constructed directly from the pollution and nightlights outcomes, it is not an independent third test. In the PM_{2.5} extension, the analogous outcome is $\log(\text{PM}_{2.5} + 0.01) - \log(\text{Nightlights} + 0.01)$.

2. Main Results

Within-DMSP rise and break. The long DMSP replication remains intact, but its internal time pattern is sharper than a simple before-and-after attenuation story. Figure 1 estimates the lag-0 birth-region coefficient separately for 5-year bins inside the DMSP-only 1992–2013 sample. The coefficient is modest in 1992–1996 (0.0085, $p = 0.472$), 1997–2001 (0.0109, $p = 0.225$), and 2002–2006 (0.0107, $p = 0.196$), then rises to 0.0292 in 2007–2011 ($p = 0.004$). In the final, shorter 2012–2013 bin, it falls to -0.0057 ($p = 0.709$). The difference between 2007–2011 and 2012–2013 is itself large and statistically significant (-0.0349 , $p = 0.005$). The basic break therefore begins within DMSP, before any switch to later nightlights products.

Pooled nested windows are consistent with this bin pattern: long DMSP windows ending in 2013 are positive (0.0128 in 1992–2013, $p = 0.030$), while pooled windows ending after the break collapse toward zero (0.0017 in 2005–2013, $p = 0.768$; 0.0004 in harmonized 2005–2019, $p = 0.966$). The later pooled null therefore reflects the late-2000s peak and early-2010s break visible in the bins, not a gradual monotonic fade.

Sub-Saharan Africa versus the rest. The historical positive effect is concentrated in Sub-Saharan Africa. Table 1 shows that in DMSP 1992–2013, the Sub-Saharan Africa coefficient

is 0.0442 ($p = 0.011$), while the non-SSA coefficient is only 0.0053 ($p = 0.370$). This is the cleanest decomposition of the original long-window result: the classic positive effect is largely an SSA phenomenon.

That historical concentration does not translate into a clean post-2005 rescue. In DMSP 2005–2013, both SSA and non-SSA are close to zero. In harmonized 2005–2019, SSA turns positive again at 0.0315, but remains imprecise ($p = 0.135$). The longer SSA-only harmonized grid shows that long windows still recover the historical pattern: the SSA 1992–2019 coefficient is 0.0448 ($p = 0.010$). The problem is therefore not that the historical SSA signal disappears everywhere. It is that short post-2005 windows do not preserve it cleanly.

Standalone VIIRS in later years. The later null is not just a DMSP-to-VIIRS splice artifact. In a standalone VIIRS-only panel, the pooled 2014–2024 lag-0 coefficient is negative at -0.0354 ($p = 0.002$), and the early VIIRS bin 2014–2016 is also negative at -0.0495 ($p = 0.001$). Standalone VIIRS therefore does not restore the canonical positive birth-region activity effect in later years.*

3. Subsample Search

If the pooled post-2005 activity reduced form masks heterogeneity, the most plausible places to look are settings where the original favoritism mechanism should be strongest or easiest to measure. We therefore test four families of subsamples: geography, political capacity, measurement headroom, and pollution-support samples. The geography cut captures Hodler and Raschky's finding of especially strong effects in Sub-Saharan Africa. The political-capacity cut isolates long-tenure leaders and long-tenure autocrats. The headroom splits ask whether DMSP saturation hides effects in already-bright districts. The pollution-support splits ask whether environmental outcomes are only informative in countries with enough NO₂ variation.

The role of this section is confirmatory rather than exploratory. No post-2005 cut restores a stable positive activity effect once the break appears. In Table 3, the other theory-favored post-2005 samples all fail to revive the reduced form: long-tenure autocrats are consistent with zero and too imprecise to distinguish from the historical effect, dropping high-luminosity districts does not help, and the high-NO₂ country sample remains near zero.

The harmonized 2005–2019 panel gives the most suggestive, though still inconclusive, later pattern. Africa (0.0262, $p = 0.167$) and Sub-Saharan Africa (0.0315, $p = 0.135$) are larger than the pooled null, but neither clears our minimum-support rule of a positive coefficient above 0.005 with $p < 0.10$. We therefore treat pollution outcomes in these samples as diagnostic rather than confirmatory (Table 4).

4. Pollution-Side Results

Pollution outcomes in the 2005–2019 sample. Pollution estimates test whether the post-2005 attenuation reflects a shift from the quantity to the composition of favored growth. In the pooled 2005–2019 sample (Table 5), the lag-0 nightlights

*In additional geography splits, the negative VIIRS-only pattern is concentrated outside Sub-Saharan Africa: the non-SSA pooled 2012–2024 coefficient is -0.0473 ($p < 0.001$), while the SSA coefficient is positive but imprecise at 0.0270 ($p = 0.378$). We do not have a convincing explanation for that later non-SSA sign reversal, so we treat it as an unresolved fact rather than a headline result.

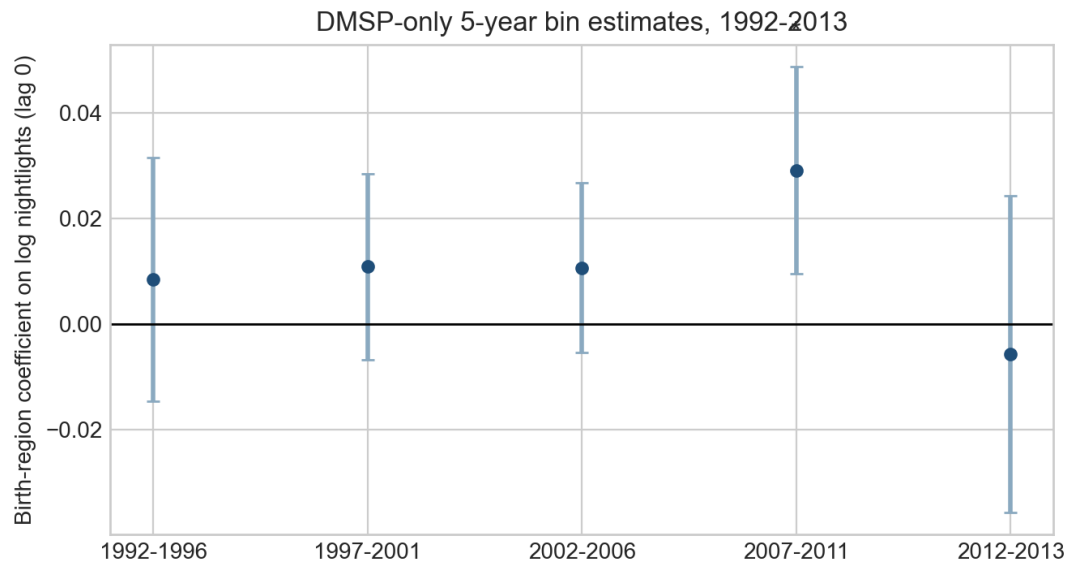


Fig. 1. DMSP-only lag-0 birth-region coefficients by 5-year bin. Each point is the coefficient from an ADM2 fixed-effect and country-year fixed-effect regression with leader-spell clustering. The final 2012–2013 bin is shorter than the earlier 5-year bins. The effect is strongest in 2007–2011 and then weakens sharply by 2012–2013.

Table 1. Sub-Saharan Africa versus non-SSA across key windows

Sample	Window	Coef.	SE	<i>p</i> -value	N	Treated
Sub-Saharan Africa	DMSP 1992–2013	0.0442	0.0174	0.011	63,378	631
Non-SSA	DMSP 1992–2013	0.0053	0.0059	0.370	888,537	1,891
Sub-Saharan Africa	DMSP 2005–2013	-0.0063	0.0164	0.700	28,191	279
Non-SSA	DMSP 2005–2013	0.0035	0.0061	0.566	370,445	789
Sub-Saharan Africa	Harmonized 2005–2019	0.0315	0.0210	0.135	52,001	466
Non-SSA	Harmonized 2005–2019	-0.0077	0.0094	0.412	619,398	1,235

All rows use lag-0 treatment, ADM2 fixed effects, country-year fixed effects, and leader-spell clustering. The long-run SSA 1992–2019 harmonized coefficient is 0.0448 ($p = 0.010$), reinforcing that the historical positive effect remains visible in long SSA windows even though the short post-2005 SSA window does not.

Table 2. Robustness to country-level clustering

Spec.	Coef.	Lead. SE	Ctry. SE	Ctry. <i>p</i>
NL, 1992–2013	0.0128	0.0059	0.0075	0.087
NL, 2005–2013	0.0017	0.0058	0.0081	0.832
NL, 2005–2019	0.0004	0.0087	0.0107	0.973
NO ₂ , 2005–2019	0.0308	0.0154	0.0204	0.132
NO ₂ int., 2005–2019	0.0333	0.0179	0.0239	0.163

NL denotes nightlights. All rows use ADM2 fixed effects and country-year fixed effects. Baseline standard errors cluster by leader spell; the robustness column clusters by country, following the original Hodler–Raschky convention.

coefficient is near zero, the lag-0 NO₂ coefficient is positive at 0.0308 ($p = 0.045$), and the lag-0 pollution-intensity outcome is also positive at 0.0333 ($p = 0.063$). These point estimates go against a cleaner-growth interpretation. At the same time, Table 2 shows that the NO₂ estimates become less precise under country clustering, so they are better read as suggestive evidence against green favoritism than as a standalone positive pollution finding.

Autocracy and PM_{2.5} extensions. The heterogeneity checks do not recover the later activity effect either. Restricting the sample to country-years with $v2x_polyarchy < 0.3$ yields a

lag-0 nightlights coefficient of 0.0111 ($p = 0.574$) in the DMSP 1992–2013 sample and 0.0076 ($p = 0.552$) in the DMSP 2005–2013 sample. These estimates are positive but too imprecise to rescue the historical result.

The PM_{2.5} extension tells the same story. In longer windows, PM_{2.5} level regressions are positive rather than negative, and in the overlapping 2005–2019 sample the PM_{2.5} pollution-intensity outcome is also not negative. The lag-2 estimate in Table 5 is positive and statistically significant, though we read it cautiously given multiple testing across lags. On balance, pollution outcomes do not explain the weaker post-2005 nightlights result as a shift toward cleaner birth-region growth.

5. Discussion

The main result is a temporal break, not just a generic later-period attenuation. Within DMSP, the birth-region activity effect is modest in early bins, strongest in the late 2000s, and then much weaker by 2012–2013. The break therefore starts before any switch to later satellite products. The historical positive effect is concentrated in Sub-Saharan Africa, but short post-2005 windows do not preserve that pattern cleanly, and standalone VIIRS does not restore it in later years.

Table 3. Targeted activity-effect subsample search

Sample	Window	Coef.	SE	p-value	N	Treated
Africa	DMSP 1992–2013	0.0428	0.0163	0.009	109,126	705
Sub-Saharan Africa	DMSP 1992–2013	0.0442	0.0174	0.011	63,378	631
Non-SSA	DMSP 1992–2013	0.0053	0.0059	0.370	888,537	1,891
Pooled	DMSP 2005–2013	0.0017	0.0058	0.768	398,636	1,068
Africa	DMSP 2005–2013	-0.0052	0.0143	0.714	46,939	306
Sub-Saharan Africa	DMSP 2005–2013	-0.0063	0.0164	0.700	28,191	279
Long-tenure leaders	DMSP 2005–2013	-0.0062	0.0087	0.479	300,372	862
Long-tenure autocrats	DMSP 2005–2013	0.0081	0.0173	0.637	42,717	261
Drop top 10% luminosity	DMSP 2005–2013	0.0010	0.0082	0.899	357,745	613
Leader birth DN < 61	DMSP 2005–2013	0.0020	0.0063	0.750	373,862	1,034
Low birth brightness	DMSP 2005–2013	0.0016	0.0108	0.880	142,320	634
High baseline NO ₂ countries	DMSP 2005–2013	-0.0012	0.0069	0.866	312,633	603
Africa	Harmonized 2005–2019	0.0262	0.0189	0.167	83,101	511
Sub-Saharan Africa	Harmonized 2005–2019	0.0315	0.0210	0.135	52,001	466
Low NO ₂ dispersion countries	Harmonized 2005–2019	0.0219	0.0183	0.232	95,961	621

All rows use lag-0 treatment, ADM2 fixed effects, country-year fixed effects, and leader-spell clustering. Long tenure means total PLAD spell length of at least five years. Long-tenure autocracy additionally requires $v2x_polyarchy < 0.3$. The minimum-support rule is coef. ≥ 0.005 , $p < 0.10$, and at least 50 treated observations in a post-2005 sample. No post-2005 subsample satisfies that rule.

Table 4. Pollution outcomes in strongest positive diagnostic samples

Sample	Activity coef.	Activity p	NO ₂ intensity	NO ₂ p	PM _{2.5} intensity	PM _{2.5} p
Sub-Saharan Africa	0.0315	0.135	-0.0323	0.237	-0.0336	0.127
Africa	0.0262	0.167	-0.0267	0.295	-0.0267	0.178
Low NO ₂ dispersion	0.0219	0.232	0.0156	0.685	-0.0203	0.281

The activity columns use the harmonized 2005–2019 nightlights panel. Pollution-intensity outcomes are log pollution minus log nightlights in the same subsample. These are diagnostic estimates because none of the activity effects is statistically significant under the stated criterion.

That temporal pattern points away from a simple harmonization story and toward changes in the underlying political or measurement environment. Several mechanisms remain plausible. One is that the late-2000s peak partly reflects the commodity-cycle period and unusually strong distributive scope in some leader spells. Another is that transparency and monitoring increased in the 2010s, making geographically concentrated favoritism harder to sustain. A third is weaker political support from shorter observed spells, though that comparison is partly mechanical in narrower windows. We also see signs of reduced luminosity headroom in later DMSP years, but the subsample tests dropping high-luminosity districts do not recover the effect. These results narrow the candidate explanations, but do not identify one of them decisively.

The pollution analysis narrows one explanation for that break. If later favoritism shifted from more growth to cleaner growth, pollution intensity should fall in leader birth regions even when nightlights do not rise. Instead, the pooled pollution estimates are zero to positive, and the negative Africa and Sub-Saharan Africa intensity point estimates in Table 4 are too imprecise to carry the cleaner-growth interpretation. The paper therefore does not solve why the nightlights reduced form breaks after the late 2000s, but it does show that the break is broad and is not explained by a simple shift toward greener birth-region favoritism.

ACKNOWLEDGMENTS. We thank Professor Joshua Blumenstock for feedback. All errors are our own.

Data, Materials, and Code Availability

Code, notebooks, and manuscript sources are available in the public GitHub repository at <https://github.com/RichSchulz/political-leader-pollution>. The reported tables are produced from the analysis notebooks in `analysis/`, including `replication.ipynb`, `green_favoritism.ipynb`, `green_favoritism_pm25.ipynb`, `final_subsample_search.ipynb`, and `final_clustering_robustness.ipynb`, together with the standalone VIIRS cache builder scripts/build_viiirs_v21_panel.py and exported diagnostics in `data/`. External inputs include PLAD, GADM, ACAG, V-Dem, harmonized nightlights data, and standalone VIIRS annual rasters obtained under the providers' access conditions.

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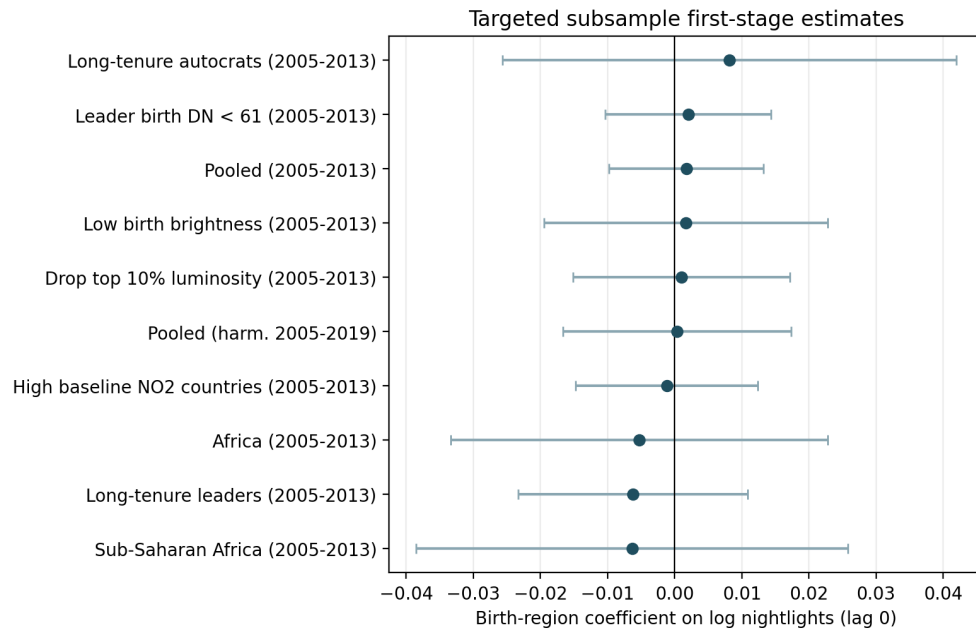


Fig. 2. Targeted post-2005 activity-effect estimates. Points are lag-0 coefficients and horizontal bars are 95% confidence intervals for pooled, Africa, Sub-Saharan Africa, long-tenure, long-tenure-autocracy, luminosity-headroom, low-brightness, and high-NO₂ country samples. The pooled harmonized 2005–2019 estimate is included as a reference row.

Table 5. Main later-period pollution results

Outcome	Window / spec	Coef.	SE	p-value	N
log(Nightlights + 0.01)	2005–2019 lag 0	-0.0026	0.0091	0.778	631,377
log(NO ₂ + 0.01)	2005–2019 lag 0	0.0308	0.0154	0.045	631,377
NO ₂ pollution intensity	2005–2019 lag 0	0.0333	0.0179	0.063	631,377
PM _{2.5} pollution intensity	2005–2019 lag 0	0.0041	0.0095	0.665	669,438
PM _{2.5} pollution intensity	2005–2019 lag 1	0.0122	0.0078	0.120	669,438
PM _{2.5} pollution intensity	2005–2019 lag 2	0.0175	0.0074	0.019	669,438

Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. Negative coefficients would indicate cleaner growth in leader birth regions; current estimates are zero to positive.